Evaluating Physics-Informed Learning for Fluid Dynamics

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Abstract

We compare a data-driven neural network (data NN), trained on simulation results, with a physics-informed neural network (PINN), trained only on the flow equations. The data NN was more accurate and cheaper; the PINN needed no data and obeyed the equations more closely.

Introduction

A trained network can stand in for a slow fluid simulation, but training has a cost: a data NN needs many simulation results, and a PINN needs more computing because every step evaluates the flow equations. We ask:

RQ1 How accurate is each network?

RQ2 How well does each obey conservation of mass and momentum?

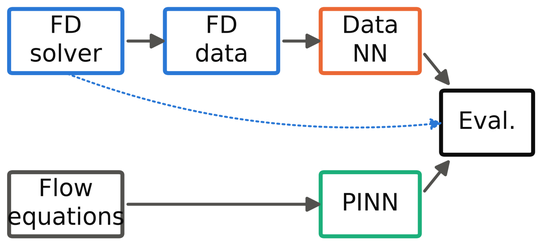
RQ3 How repeatable is each across random seeds and flow speeds?

RQ4 What does each cost to train and run?

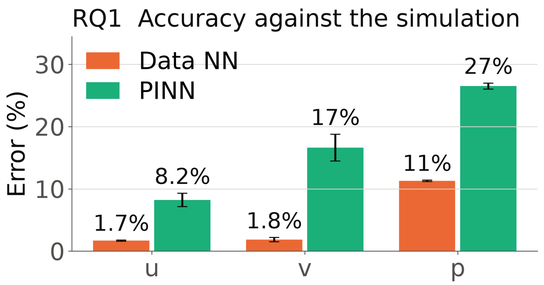
Methods

Fluid in a square box is driven by a sliding lid (Reynolds number Re = 100 and 400). A finite-difference (FD) simulation is the reference and matches published results [1, 2].

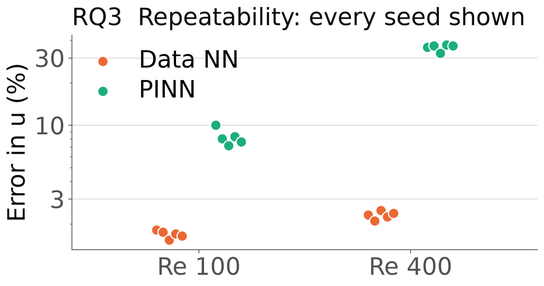
Both networks have 17,091 parameters, train for 20,000 steps with 5 seeds, and are scored by one evaluator on data never used in training.



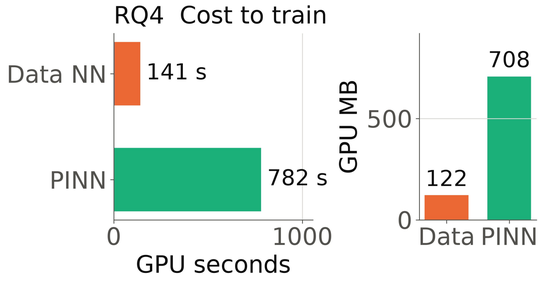
Results



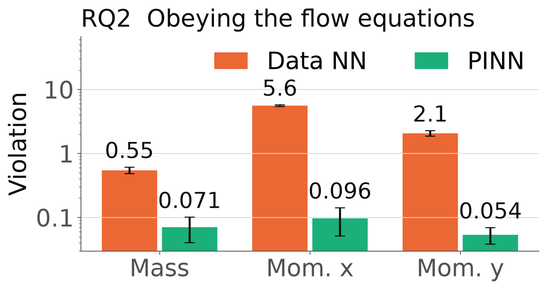
The data NN's error is 1.7% in u and 11% in p; the PINN's is 8.2% and 27% (Re = 100, mean of 5 seeds).



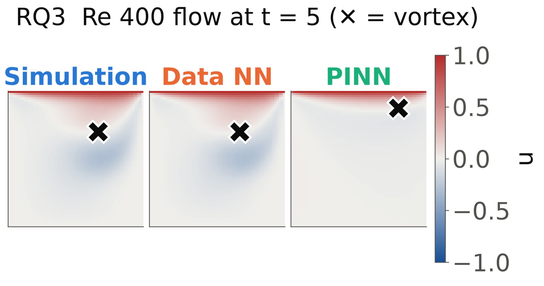
Both networks repeat well across seeds at Re = 100. At Re = 400 the data NN stays near 2.3%, but every PINN seed fails (36%).



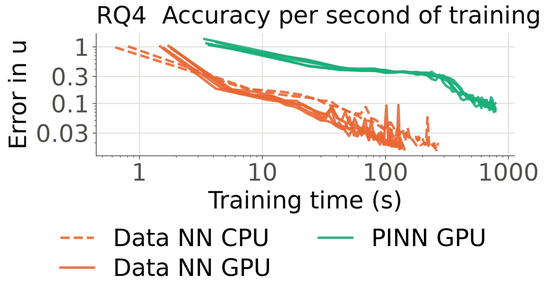
The data NN needs 3.9 s of simulation plus 137 s of training; the PINN needs 782 s and more GPU memory.



The PINN obeys all three equations more closely: mass violation 0.071 vs 0.55 for the data NN.



The failed PINN still conserves mass (violation 0.03 vs 0.55), yet its vortex sits by the lid: obeying the equations is not the same as being right.



Error in u during training (solid GPU, dashed CPU). Once trained, both return the whole flow in about 10 ms.

Strengths & weaknesses

	Data NN	PINN
Needs simulation data	Yes	No
Error in u, Re 100	1.7%	8.2%
Error in u, Re 400	2.3%	36%
Mass violation	0.55	0.071
GPU training	137 s	782 s
GPU memory	122 MB	708 MB
CPU time per step	13 ms	297 ms

Green marks the better of the two (Re = 100 unless stated).

Conclusion

Neither approach wins on every measure. The data NN gave the closest answers for the least computing, but only because simulation data already existed. The PINN needed no data and obeyed the equations better, but cost more to train and failed at the faster flow.

Their strengths and weaknesses are opposite. This points toward a hybrid that adds a small amount of data to the physics loss, which could reduce both the data and the computing needed. A hybrid was not tested here; it is the next step.

References

[1] Marchi, C. H., Suero, R., and Araki, L. K. 2009. The lid-driven square cavity flow: numerical solution with a 1024 x 1024 grid. J. Braz. Soc. Mech. Sci. Eng. 31, 3, 186–198.

[2] Hou, S., Zou, Q., Chen, S., Doolen, G., and Cogley, A. C. 1995. Simulation of cavity flow by the lattice Boltzmann method. J. Comput. Phys. 118, 2, 329–347.

[3] Raissi, M., Perdikaris, P., and Karniadakis, G. E. 2019. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. J. Comput. Phys. 378, 686–707.

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